



Characteristics and controls of solute transport under different conditions of soil texture and vegetation type in the water–wind erosion crisscross region of China's Loess Plateau

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HIGHLIGHTS

- We studied Cl^- transport under different soil textures and vegetation types.
- Both soil texture and vegetation type greatly affect the behavior of Cl^- transport.
- Pedotransfer functions can adequately simulate and predict Cl^- transport parameter.

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ABSTRACT

The analysis of solute transport characteristics in soil is of great significance in understanding nutrient cycling and pollutant migration in the Earth's Critical Zone. The objective of this study was to investigate the transport characteristics and the influencing factors of Cl^- in soils with different textures (sandy-S and loamy-L), and covered by different vegetation types (arbor-AR, shrub-SH and grass-GR) in the water–wind erosion crisscross region of the northern Loess Plateau of China. Results showed that the initial penetration time (TS: 12–80 min), entire penetration time (TE: 75–480 min), average flow velocity in the pore (V : 0.52–1.98 cm h^{-1}) and the hydrodynamic diffusion coefficient (D : 0.75–2.55 $\text{cm}^2 \text{h}^{-1}$) of Cl^- varied with different soil textures and vegetation types, and at different soil depths. The V and D associated with Cl^- transport were highest in the 0–20 cm soil layer and decreased with increasing depth, while the opposite trend was observed for TS and TE. For the 0–1 m soil profile of the same texture but covered by different vegetation types, the average V and D followed the order of S-AR > S-GR > S-SH and L-AR > L-SH > L-GR, while the average TS and TE exhibited the exact opposite order. This behavior is caused by the varying distributions of root biomass under different vegetation types that affect the number of macropores, the connectivity density and the preferential flow paths in the soil. For the 0–1 m soil profiles of different textures covered by the same vegetation type, the average V and D followed the order of S-AR > L-AR; S-SH > L-SH; and S-GR > L-GR, while the average TS and TE showed the opposite trend. This is because the pore size and distribution in soil are significantly affected by soil mechanical composition. There are significant correlations between soil properties (e.g., bulk density, number of macropores, pore connectivity density, saturated hydraulic conductivity, soil organic carbon content and particle composition) and the transport parameters (e.g., V , TS, and TE). The pedotransfer functions using readily available soil properties can adequately predict V of Cl^- transport under different conditions of soil texture and vegetation type. These results provide guidance for the rational configuration of artificial vegetation in different textural soils with respect to reduce nutrient loss and improve ecosystem functions in the northern Loess Plateau of China.

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1. Introduction

The transport of solute in porous media is affected by physico-chemical processes, biological activities and human intervention (Jarvis, 2007; Zhen et al., 2019). Analyzing the process, pattern, and mechanism of solute transport in soil is of great significance in understanding nutrient cycling and pollutant migration in the Earth's critical zone. Studying this will also assist in formulating better management strategies in order to reduce nutrient loss and minimize the migration of pollutants to water bodies. Therefore, the transport behavior of solute in soil has extensively been researched, and a number of objectives have been achieved: (i) the establishment of a solute transport model in soil at different scales (Van Genuchten et al., 1989; Larsson and Jarvis, 1999; Zhou et al., 2019); (ii) identifying the differences in solute transport in homogeneous and heterogeneous (e.g., multilayered) soil profiles, and the corresponding influencing factors (Porro et al., 1993; Wang et al., 2007); (iii) understanding the mechanism and management strategy of solute transport under different flow velocities and pore conditions (Costa and Prunty, 2006; Hu et al., 2009; Zhao et al., 2019); and (iv) the application of the solute transport model to improve saline and alkali soils, the management of soil nutrients, and the evaluation of nonpoint source pollution (Li et al., 2005). Previous studies have shown that the transport behavior of solute in soil is affected by a combination of factors such as the physical and chemical properties of the soil (Zhen et al., 2015), the flow characteristics of water (Guo et al., 2013), type of vegetation cover (Min et al., 2018), biological activities (Edwards et al., 1993), topography (Delin and Landon, 2002), and human management practices (Okada et al., 2014). Solute transport parameters must be accurately estimated before analyzing the transport behavior of soluble substances in a porous medium. These parameters also play a fundamental role in the evaluation and prediction of the temporal and spatial distribution of nutrients and pollutants. In recent years, the breakthrough curve (BTC) method (Yamaguchi et al., 1989), the boundary layer method (Shao et al., 1998), and the polynomial method (Wang et al., 2017) have been widely used to estimate the parameters in the convection–diffusion equation (CDE). These methods provide important means for analyzing the migration and transformation of agrochemicals, nutrients, or pollutants in soil quantitatively.

The border area between Shanxi, Shaanxi, and Inner Mongolia, located in the northern Loess Plateau, is a typical water–wind erosion crisscross region and an ecologically fragile area. Since the implementation of the “Grain for Green Program” in 1999, the vegetation cover in this region has increased significantly, providing effect measures to combat soil erosion and desertification (Deng et al., 2012). Previous studies showed that the process of vegetation restoration and ecological reconstruction in the Loess Plateau had an important impact on the evolution of soil properties (Zhang et al., 2020), hydrological processes (Huang and Shao, 2019), nutrient cycling (Wang et al., 2001), and solute transport behavior (Turkeltaub et al., 2018). Studies on the solute transport in soil in this region have primarily focused on the following three aspects: (i) the transport characteristics of solute in deep loess and the corresponding influencing factors (e.g., Zhou and Wang, 2017; Zhen et al., 2015, 2016, 2019; Wang and Shao, 2018); (ii) the application of the solute transport model (e.g., two-flow region model) and the methods for estimating the model parameters (e.g., Yang et al., 2013; 2016a; 2016b; 2016c); and (iii) the enrichment and loss of nutrients/pollutants in soil and the corresponding influencing factors (e.g., Pan et al., 2013; Tao et al., 2020). These studies have provided an in-depth understanding of the solute transport process in typical loess areas, as well as the primary controlling factors. In addition, the results from these studies have assisted in formulating

more reasonable management strategies to reduce nutrient loss in soil and in minimizing the migration of pollutants to water bodies. However, it remains unclear how solute transport parameters and the transport characteristics are affected by different soil properties associated with different soil textures and vegetation types in the Loess Plateau. Without such an understanding, developing accurate models of nutrient cycling and pollutant migration will remain a challenge. In this study, the transport behavior of solute in soil associated with different soil textures (sandy-S and loamy-L) and vegetation types (arbor-AR, shrub-SH, and grass-GR) in the Liudaogou and Gechougou watersheds in the water–wind erosion crisscross region of the northern Loess Plateau was analyzed by estimating the solute transport parameters from the BTC of the flux concentration. The primary factors that affect the solute transport in soil are discussed in detail. These results provide a basis to the formulation of a more reasonable layout and configuration of artificial vegetation for different soil textures, the reduction of nutrient loss risk, and the improvement of ecosystem service and function.

2. Materials and methods

2.1. Overview of the study area

The region investigated in this study include the Liudaogou watershed (38°46′–38°51′ N, 110°21′–110°23′ E) and Gechougou watershed (38°53′–38°57′ N, 109°22′–109°27′ E) in Shenmu, Shaanxi Province (Fig. 1). The two watersheds are located in the transition zone between the Loess Plateau and the Mu Us Sandy Land, representing a typical water–wind erosion crisscross region and an ecologically fragile area (Fig. 1). Soils in the two watersheds have contrasting textures, which provide an excellent opportunity to study solute transport characteristics in different textural soils. To alleviate severe soil erosion and desertification, vegetation restoration has been widely implemented in the studied watersheds since 1998.

The Liudaogou watershed is located approximately 14 km west of Shenmu, with an average altitude of 1178 m, an annual average temperature of 8.4 °C, and an annual average precipitation of 442 mm (1971–2013). Rainfall from July to September accounts for 70%–80% of the annual precipitation. The main soil types are Calcic Aridisols and Haplic Aridisols according to Chinese Soil Taxonomy (Gong et al., 2007) or Haplocalcids and Haplocambids according to Keys to Soil Taxonomy (Soil Survey Staff, 2010), which develops from low-fertility loess. The restored vegetation typical in the watershed includes purple alfalfa (*Medicago sativa*), korshinsk peashrub (*Caragana korshinskii* kom (*C. korshinskii*)) and apricot trees (*Prunus armeniaca* (*P. armeniaca*)). The abandoned cropland in the studied watershed is generally recovered by natural vegetation, such as bunge needlegrass (*Stipa bungeana* Trin (*S. bungeana*)) and dahurica bush clover (*Lespedeza dahurica*) (Mao et al., 2018).

The Gechougou watershed is located approximately 23 km southwest of Jinjie Town, Shenmu County. It has an average altitude of 1265 m, an annual average temperature of 9.1 °C, and an annual average precipitation of 420 mm (2006–2018). Approximately 60%–75% of the annual rainfall occurs from July to September. The terrain of the Gechougou watershed is fairly flat. Most of the sand dunes are relatively short and distributed along a belt track. The main soil type is Aridi-Sandic Primosols according to Chinese Soil Taxonomy (Gong et al., 2007) or Xeropsamments according to Keys to Soil Taxonomy (Soil Survey Staff, 2010). Artificial vegetation has been widely introduced into this catchment to control severe soil erosion and degradation during past several decades. The typical

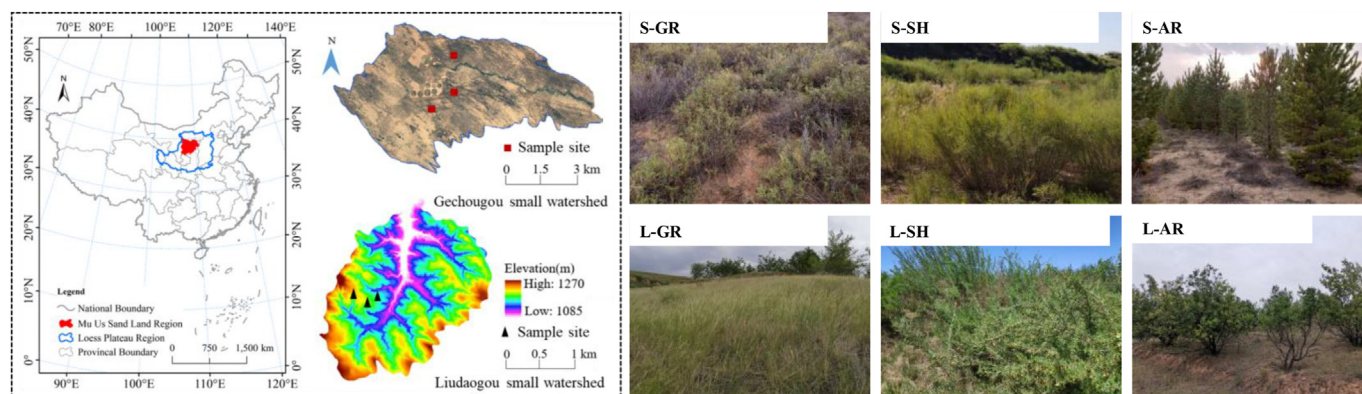


Fig. 1. Landscape with different vegetation types growing in sandy soil and loamy soil in the Gechougou and Liudaogou watersheds, respectively. S-AR, S-SH, and S-GR represent *P. sylvestris*, *S. psammophila*, and *A. destororum* in sandy soils in the Gechougou watershed, respectively; and L-AR, L-SH, and L-GR represent *P. armeniaca*, *C. korshinskii*, and *S. bungeana* in loamy soils in the Liudaogou watershed, respectively.

plants used in vegetation restoration in the region include arbor (e.g., *Pinus sylvestris* var. *mongolica* Litv. (*P. sylvestris*)), shrub (e.g., *Salix psammophila* (*S. psammophila*), *Amorpha fruticosa* Linn, and *Amygdalus pcdunculata* Pall), grasses (e.g., *Artemisia destororum* Spreng (*A. destororum*), *Setaria viridis*, and *Stipa bungeana* Trin) (Pei et al., 2019).

2.2. Sample collection and analysis

The soil samples were obtained from 1.2 m deep soil profiles excavated from typical arbor land, shrubland, and grassland in the Liudaogou and Gechougou watersheds. The three loamy soil profiles taken from the Liudaogou watershed were covered by *P. armeniaca*, *C. korshinskii* and *S. bungeana*, respectively. These sample plots were labeled L-AR, L-SH, and L-GR (Fig. 1). The three sandy soil profiles taken from the Gechougou watershed were covered by *P. sylvestris*, *S. psammophila*, and *A. destororum*, respectively. These sample plots were labeled S-AR, S-SH, and S-GR, respectively (Fig. 1). After identifying different layers in the soil profile on the site, the undisturbed soil columns were collected using plexiglass columns (height (H) = 25 cm, diameter (D) = 7 cm) with a sampling interval of 20 cm. During sample collection, a 5 cm void space was left in the top section of each plexiglass column. Due to the presence of a large number of calcium carbonate nodules in the soil below 1 m depth, the undisturbed soil columns were sampled to 1 m deep in the Liudaogou watershed. The sampling process was repeated three times in each layer. The collected undisturbed soil columns were used for computed tomography (CT) scanning and solute transport experiment. In order to analyze the basic physical and chemical properties of the soil, additional undisturbed and disturbed soil samples were collected using stainless steel cutting rings (H = 5 cm, D = 5 cm) and a cloth bag. The sampling intervals were 10 cm and 20 cm at a depth within 1 m and beyond 1 m, respectively. To analyze the growth condition of the above-ground plants, vegetation samples were collected from a 10 m × 10 m, 5 m × 5 m, and 2 m × 2 m plot in arbor land, shrubland, and grassland, respectively. The sampling process was repeated three times in each sampling plot. Plant roots were collected with a root drill (H = 10 cm, D = 9 cm). The sampling interval was set as 10 cm at a sampling depth within 60 cm, and 20 cm at a sampling depth between 60 cm and 100 cm.

Undisturbed soil samples collected by stainless steel cutting rings were heated in an oven at 105 °C for measuring the bulk density (BD) and soil water content (SWC). The saturated hydraulic conductivity (K_s) was measured using the constant hydraulic head

method (Jury and Horton, 2004). Disturbed soil samples were brought back to the laboratory and air-dried. After removal of litter, roots and gravel, the soil samples were ground and screened through 2 mm, 1 mm, and 0.25 mm sieves for determining soil organic carbon (SOC), total phosphorus (TP) and particle size composition. SOC was measured using the $K_2Cr_2O_7$ oxidation method (Nelson et al., 1982). TP was measured the ascorbic acid molybdenum blue method (Murphy and Riley, 1962). The composition of the soil particles was measured using an MS2000 laser particle size analyzer (Malvern Instrument, Malvern, England) and subsequently categorized into clay particles (<0.002 mm), powder particles (0.02–0.002 mm), and sand particles (2–0.02 mm) according to the international texture classification system. Detailed low-noise projections of undisturbed soil columns at a length scale of 1 mm were obtained using a Philips MX 16-Slice CT scanner (Brilliance16 & MX16). The scanning parameters of the instrument were set as 140 kV, 200 mA, and 21.4 s. Both the width of the X-ray and the thickness of the soil slice were set as 1 mm. The pore-related parameters of the undisturbed soil columns were further extracted from the X-ray image using ImageJ software (version 1.48). Specifically, the target area in the X-ray image was first selected using the oval selection tool. The regions outside the target area were then deleted from the image to eliminate the influence of the plexiglass column wall on the surrounding soil as well as to reduce interference from the hardening beam. It should be noted that substances with different densities in the plexiglass column will show different gray values in the image. Therefore, a threshold gray value was set in this study to separate substances with different densities in the soil. In particular, the parts with a gray value greater than 40 were considered as the porous space in the soil, whereas the remaining parts were considered as non-porous space such as the soil substrate, rock particles, and live roots. Due to limitations in resolution of the CT scan image, all the pores identified in ImageJ were considered as large pores. The detailed analysis procedures can be found in Li et al. (2018).

2.3. Solute transport experiment

The solute transport experiment was conducted in the laboratory using a plexiglass column (H = 25 cm, D = 7 cm). The top part of the soil column was a water supply port, and the bottom part was a solution leaching outlet made from porous glass (Fig. 2). During the experiment, filter paper (1–3 μm) was placed at the water injection port to prevent the structure of the soil in the top region being damaged by the injected water. At the same time, a gauze

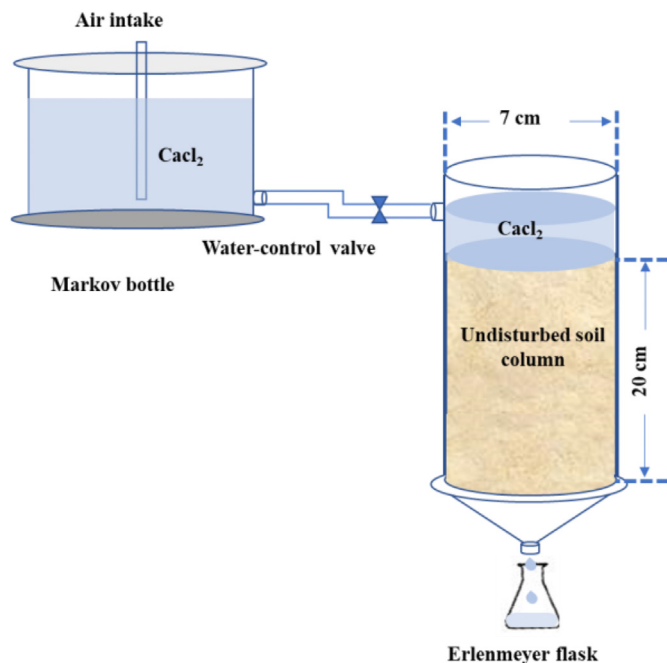


Fig. 2. Schematic of Cl^- transport experiment.

cloth was placed above the lower porous glass to prevent leakage of soil particles from the plexiglass column (Fig. 2). During the test, the water was supplied using a Mariotte bottle to ensure a constant head (at approximately 3 cm). After the soil column was fully saturated by distilled water and the outflow rate became stable, CaCl_2 test solution with a concentration of 0.5 mol l^{-1} was injected from the upper of the column into the soil sample. Once the test started, the volume of the liquid flowing out of the soil was measured every 5 and 30 min for the sandy and loamy soils, respectively, by collecting the outflow liquid in a 25 mL volumetric flask. The concentration of Cl^- in the outflow solution was measured by 0.1 mol l^{-1} silver nitrate titration (Wang and Shao, 2018). The initial penetration time and the entire penetration time were recorded as TS and TE, respectively.

2.4. Theoretical basis of the CDE equation

The variation in the relative concentration of the outflow liquid with the pore volume is described by the BTC. The shape of the BTC not only reflects the interaction between the solute and the soil, but also directly indicates how the solute concentration changes during the transport process. The BTC of soil associated with different soil textures and vegetation types can be plotted using the CDE equation. In this way, the solute transport parameters can be obtained using the CXTFIT software package. The solute transport process was analyzed based on the convection–diffusion equation:

$$R \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} + V \frac{\partial C}{\partial x} \quad (1)$$

The BTC can be used to provide a quantitative relationship between the relative concentration (C/C_0) of solute and the corresponding time. Here, C_0 is the concentration of the solute (i.e., Cl^- in this study) in the incoming liquid, C is the concentration of the solute in the outgoing liquid, L is the length of the soil column, R is the retardation factor, t is the flow time, V is the average flow velocity in the porous medium, x is the flow distance, and D is the hydrodynamic diffusion coefficient. The diffusivity λ and the Péclet

number Pe can be calculated based on D , V , and L as follows:

$$\lambda = \frac{D}{V} \quad (2)$$

$$Pe = \frac{LV}{D} = \frac{1}{\lambda} \quad (3)$$

3. Results

3.1. Soil physical and chemical properties

Fig. 3 shows the variations in BD, SOC, TP, K_s , clay, silt, sand, and SWC of the samples associated with different soil textures and vegetation types in the study area. As shown in Fig. 3, the coefficients of variation of the BD in different soil profiles (S-AR, S-SH, S-GR, L-AR, L-SH and L-GR) were all smaller than 10% (2.5%–6.9%), which can be considered as a weak variation. The average BD of the sandy soils (S-AR, S-SH, and S-GR) was significantly higher than that of the loamy soils (L-AR, L-SH, and L-GR) (Fig. 3) ($p < 0.05$), which is consistent with the result of a previous study conducted by Wang et al. (2014) on the entire Loess Plateau. SOC content of the different soil samples (S-AR, S-SH, S-GR, L-AR, L-SH and L-GR) varied between 0.61 and 6.35 g kg^{-1} , which was close to that measured in the Zhifanggou watershed in the semi-arid region of the Loess Plateau (1.91 – 6.72 g kg^{-1}) (Zhao et al., 2016), but substantially smaller than that measured in the semi-humid region in the southeastern Loess Plateau (4.82 – 11.9 g kg^{-1}) (Zhao et al., 2015). The average SOC content of the S-SH and S-GR samples was significantly smaller than that of L-SH and L-GR ($p < 0.05$), while there was no significant difference in the average SOC content of S-AR and L-AR (Fig. 3). The difference in SOC content between forestland and grassland at the slope scale are attributed to the differences in surface litter, root exudates, and microbial biomass (Mao et al., 2018). TP content of the different soil samples (S-AR, S-SH, S-GR, L-AR, L-SH and L-GR) varied between 0.14 and 0.59 g kg^{-1} , which was close to that measured for different land uses in the Loess Plateau (0.27 – 0.38 g kg^{-1}) (Liu et al., 2013).

The average K_s of the sandy soils (S-AR, S-SH, and S-GR) measured 0.51 , 0.35 , and 0.38 cm min^{-1} , respectively, which were significantly higher than the average K_s ($\sim 0.03 \text{ cm min}^{-1}$) of the loamy soils (L-AR, L-SH, and L-GR) ($p < 0.05$) (Fig. 3). A study by Ahmad et al. (2016) in arid and semi-arid saline–alkali land also showed that the K_s of sandy soils is significantly higher than that of loamy soils. The average sand content in the S-AR, S-SH and S-GR (93.1%–97.9%) was significantly higher than that in the L-AR, L-SH and L-GR (15.8%–59.5%). In contrast, the average silt and clay contents in the S-AR, S-SH and S-GR (2.7%–9.0%) were substantially lower than those in the L-AR, L-SH, and L-GR (38.8%–83.1%) (Fig. 3). SWC of the sandy soils (S-AR, S-SH, and S-GR) varied between 1.34% and 3.13% (Fig. 3), which is close to that measured at the southeastern edge of Mu Us Sandy Land (2.17%–3.57%) (Nan et al., 2020). SWC of the loamy soils (L-AR, L-SH, and L-GR) (4.84%–11.49%) was substantially higher ($p < 0.05$) than that of the sandy soils (Fig. 3), but close to that measured by Cao et al. (2020) (5.80%–18.50%) in the Liudaogou watershed in the Loess Plateau.

Table 1 shows the statistical analysis on the number of macropores, the area ratio of the macropores (i.e., the ratio of macropores area to the total soil area), and the pore connectivity density of the soil samples with different soil textures and vegetation types. The average number of macropores in the sandy soils (S-AR, S-SH, and S-GR) was 18, 13, and 14, respectively, which was greater than the average number of macropores found in the loamy soils (L-AR, L-

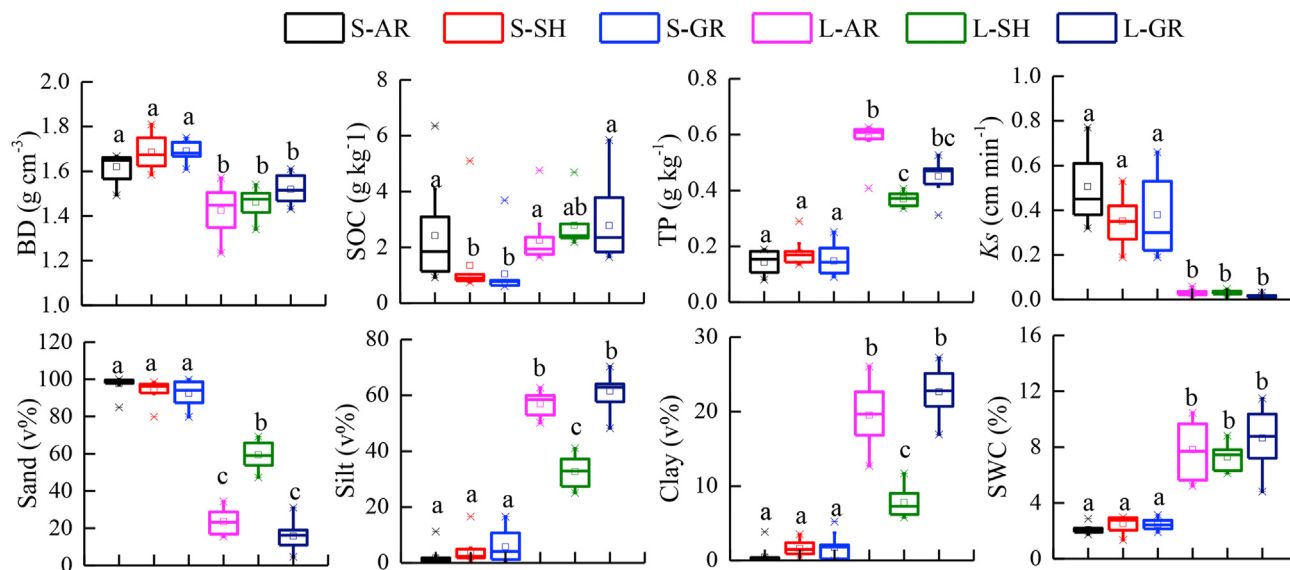


Fig. 3. Variation of soil bulk density (BD), soil organic carbon (SOC), total phosphorus (TP), saturated hydraulic conductivity (K_s), soil water content (SWC), and sand, silt, and clay content in the studied soils. S-AR, S-SH, and S-GR represent *P. sylvestris*, *S. psammophila*, and *A. dexterorum* in sandy soils in the Gechougou watershed, respectively; and L-AR, L-SH, and L-GR represent *P. armeniaca*, *C. korshinskii* and *S. bungeana* in loamy soils in the Liudaogou watershed, respectively. Lowercase letters indicate significant differences at a level of 0.05.

Table 1

Descriptive statistics of macropore parameters (number, ratio in area, and connectivity density) in the studied soils.

Plot ^a	Number of macropores ^b				Ratio in area of macropores				Connectivity density (cm^{-3})			
	Min	Max	Average	SD	Min	Max	Average	SD	Min	Max	Average	SD
S-AR	10.0	28.0	18.0a	7.1	2.4	5.7	3.8a	1.3	1.0	5	2.8a	1.8
S-SH	7.0	19.0	13.0b	4.5	1.4	3.9	2.6b	0.9	0.8	3.1	1.6b	0.9
S-GR	8.0	24.0	14.0 ab	7.0	1.6	4.9	2.8 ab	1.4	0.27	3.6	1.4b	1.4
L-AR	8.0	22.0	14.0ac	5.3	1.6	4.5	3.0 ab	1.1	0.32	2.6	1.2b	0.9
L-SH	5.0	18.0	11.0c	4.8	1.0	3.7	2.4bc	1.0	0.09	1.7	0.6c	0.7
L-GR	2.0	12.0	6.0d	3.8	0.4	2.4	1.2d	0.8	0.06	0.4	0.2d	0.1

^a S-AR, S-SH, and S-GR represent *P. sylvestris*, *S. psammophila*, and *A. dexterorum* in sandy soils in the Gechougou watershed, respectively; and L-AR, L-SH, and L-GR represent *P. armeniaca*, *C. korshinskii* and *S. bungeana* in loamy soils in the Liudaogou watershed, respectively.

^b Min, Max and SD represent minimum, maximum and standard deviation, respectively; lowercase letters in the columns of "Average" indicate significant differences at a level of 0.05.

SH, and L-GR) (14, 12, and 6, respectively) (Table 1). The variation in the area ratio of macropores was found to follow a similar trend with the differences in the number of macropores (Table 1). Both the number and area ratio of macropores in the soil were two dimensional parameters of the porous space, and the pore connectivity density was regarded as a three-dimensional parameter to better describe the characteristics of the porous space in the soil. As shown in Table 1, the connectivity density of the sandy soils (S-AR, S-SH, and S-GR) was greater than that of the loamy soils (L-AR, L-SH, and L-GR). In addition, the connectivity density of the soil samples with the same texture was found to be higher in arbor land (S-AR and L-AR) than those in shrubland (S-SH and L-SH) or grassland (S-GR and L-GR). A strong connectivity between the pores can improve the infiltration capacity of water into the soil while reducing surface runoff (Gryze et al., 2006). Therefore, both soil texture and vegetation type have a substantial impact on the characteristics of pores in the soil.

3.2. Breakthrough curve and transport parameters of Cl^-

Fig. 4 shows the BTC, TS, and TE of Cl^- transport in undisturbed soil columns at depths of 0–20, 20–40, 40–60, 60–80, and 80–100 cm from the plots with different soil textures and

vegetation types. The surface soil (0–20 cm) of the different plots (S-AR, S-SH, S-GR, L-AR, L-SH and L-GR) had the smallest TS and TE. With increasing depth, TS and TE became greater (Fig. 4), suggesting that Cl^- transport experiences less resistance in the surface soil. This is attributed to the high porosity caused by the abundant roots in the surface soil. TS and TE measured in the loamy soils at the different depths were all greater than those measured in the sandy soils covered by the same vegetation type (L-AR > S-AR; L-SH > S-SH; and L-GR > S-GR) (Fig. 4). This is because the pore distribution in the soil under the same vegetation type is determined by its mechanical composition. The sandy soil was mainly composed of coarse-grained sand (Fig. 3), resulting in large pores along with a high connectivity density (Table 1). This is conducive to the formation of preferential flow so that the solute experiences less retardation during the transport process in the sandy soil (Zhen et al., 2016). In contrast, the silt and clay contents in the loamy soils were both higher than those in the sandy soils (Fig. 3). The adsorption of ions onto charged aggregates and the complex porous geometry formed by the fine pores in the internal structure of the loamy soil will both hinder the solute transport, thereby resulting in a high TS and TE (Siyal et al., 2013).

As shown in Fig. 4, soil samples with the same texture but covered by different vegetation types have different BTC shapes,

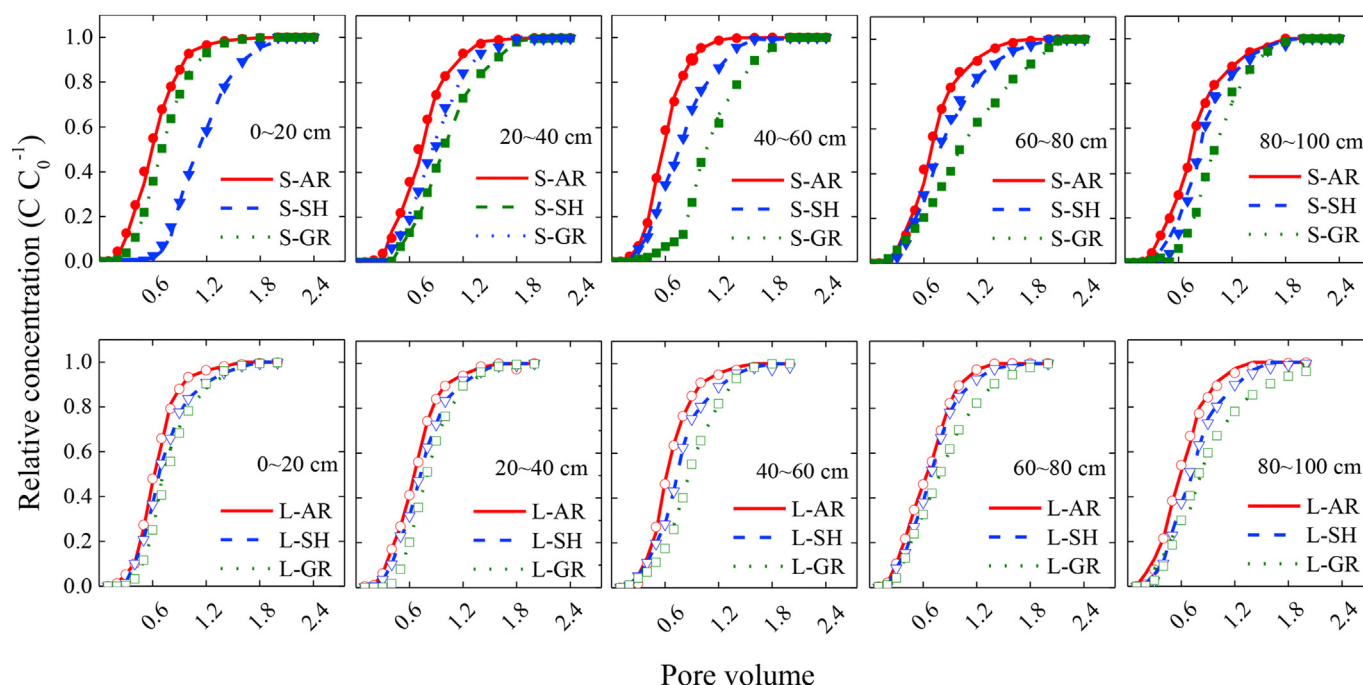


Fig. 4. Cl^- penetration curve in the studied soils. C/C_0 in the Y-axis means the ratio of Cl^- concentration in the outgoing liquid (C) to Cl^- concentration in the incoming liquid (C_0). S-AR, S-SH, and S-GR represent *P. sylvestris*, *S. psammophila*, and *A. desertorum* in sandy soils in the Gechougou watershed, respectively; and L-AR, L-SH, and L-GR represent *P. armeniaca*, *C. korshinskii* and *S. bungeana* in loamy soils in the Liudaogou watershed, respectively.

and different TS and TE. TS and TE in the sandy soils followed the order of S-AR < S-SH < S-GR at the depth of 0–40 cm and the order of S-AR < S-SH < S-GR at the depth of 40–100 cm. TS and TE measured in the loamy soils, however, followed the order of L-AR < L-SH < L-GR regardless of the soil depth. This is primarily related to the distribution characteristics of the roots of different types of vegetation. The root biomass in the sandy soils followed the order of S-AR (277.4 g m^{-2}) > S-SH (112.6 g m^{-2}) > S-GR (40.1 g m^{-2}) at the depth of 0–40 cm and the order of S-AR (84.5 g m^{-2}) > S-SH (77.2 g m^{-2}) > S-GR (40.1 g m^{-2}) at the depth of 40–100 cm. The root biomass in the loamy soils followed the order of L-AR > L-SH > L-GR at different soil depths. The total root biomass in the 0–100 cm soil profile in plots of L-AR, L-SH and L-GR were 422.5, 370.2, and 196.5 g m^{-2} , respectively. This suggests that the mechanical penetration of roots into the soil has a significant impact on the number and connectivity density of pores in the soil. The roots of arbor trees are generally stronger and distributed more widely in soils with varying textures, which creates a greater number of macropores as well as a larger connectivity density in the soil (Table 1). Therefore, TS and TE are relatively smaller in the soils covered by arbor; the peak value of the corresponding BTC also shifted slightly to the left (Fig. 4). The root biomass of the sandy soils at the depth of 0–40 cm was found to be greater when covered by grassland compared to that covered by shrubland. This is possibly caused by the low SWC at the surface of the sandy soil which impedes the growth of shrub roots. A general flattening trend was observed in all the BTCs of the soil samples, indicating that the transport and diffusion of solute in the soil become slower over time; consistent with the results obtained by Zhen et al. (2016).

The solute transport parameters can provide a quantitative description of the solute transport process and the corresponding variation trend. Fig. 5 shows the transport parameters of Cl^- in soils with different textures and vegetation types (S-AR, S-SH, S-GR, L-AR, L-SH and L-GR). The residual sum of squares and the coefficient

of determination (R^2) of the fitting curve were between 0.0002 and 0.0063, and 0.93 and 0.99, respectively. The average flow velocity V of Cl^- varied in soils under different vegetation types (Fig. 5), which is consistent with the findings obtained by Zhen et al. (2015) in a reclaimed area of a coal mine. As demonstrated in a study by Zhen et al. (2015), soils covered by different vegetation types possess different soil structures and pores characteristics due to the distinctive root distribution of the different vegetation types. Therefore, the preferential flow paths will vary for soils covered by different types of vegetation. However, as mentioned earlier, the growth of different plant roots is affected by a combination of factors including soil texture, nutrient conditions, and SWC. Therefore, the variation in average flow velocity V in the sandy and loamy soils does not follow the same pattern for different vegetation types (Fig. 5). When covered by the same type of vegetation, the average flow velocity V of Cl^- was found to be higher in the sandy soil than in the loamy soil (S-AR > L-AR; S-SH > L-SH; and S-GR > L-GR) (Fig. 5). This behavior is attributed to the coarser texture of sandy soil (Fig. 3), which yields a greater number of macro-pores as well as a better pore connectivity than loamy soil (Table 1). As a result, the solute transport experiences less resistance in the sandy soil than in the loamy soil. Liu and Xu (2005) analyzed the process of solute transport in soils with different textures and also found that the solute transport velocity V was higher in sandy soil than in loamy soil. The hydrodynamic diffusion coefficient D provides a comprehensive characterization of the solute and the soil. This parameter is affected not only by the properties of the porous medium and the solute, but also by the flow velocity V . The variations in D and V were found to follow the same pattern among soil samples with the same texture but covered by different vegetation, or with different textures but covered by the same vegetation. Past studies have shown that D is proportional to the square of V (Singh and Kanwar, 1991; Wang et al., 2018). The same relationship was also observed in this study. The Péclet number Pe in soils with different textures and vegetation types (S-AR, S-SH, S-GR, L-AR, L-

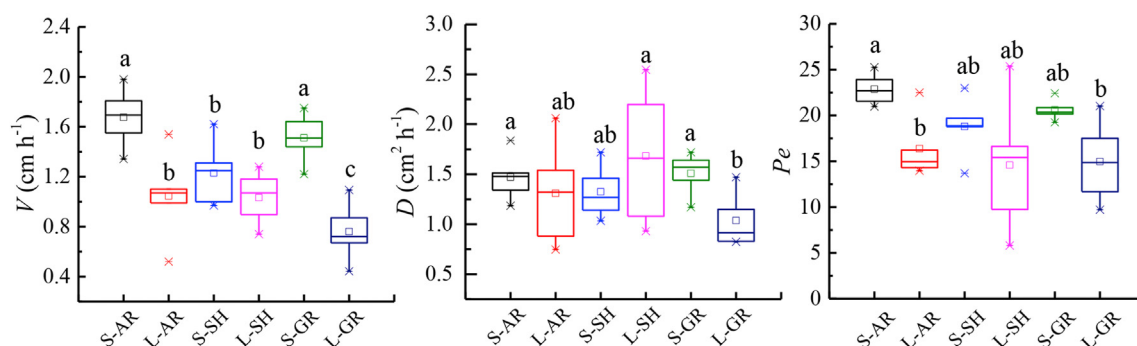


Fig. 5. Cl^- transport parameters in the studied soils. Note: S-AR, S-SH, and S-GR represent *P. sylvestris*, *S. psammophila*, and *A. desterorum* in sandy soils in the Gechougou watershed, respectively; and L-AR, L-SH, and L-GR represent *P. armeniaca*, *C. korshinskii* and *S. bungeana* in loamy soils in the Liudaogou watershed, respectively. Lowercase letters indicate significant differences at a level of 0.05.

SH, and L-GR) varied within the ranges of 20.97–25.27, 13.70 to 22.98, 19.24 to 20.85, 13.96 to 22.49, 5.81 to 25.38, and 9.70 to 21.02, respectively. The Péclet number measures the relative importance of convective flux to the diffusive flux. When Pe is greater than 1, the contribution from the convective transport is greater than that from the diffusion transport in the total mass flux (Raats, 1973). Therefore, convection was the primary mode of solute transport in this study.

3.3. Influencing factors of Cl^- transport parameters

Table 2 shows the relationships between soil properties and Cl^- transport parameters. As shown in Table 2, BD was significantly negatively correlated to V and D , but significantly positively correlated to TS and TE. On the other hand, K_s , the number of macropores, and the connectivity density were significantly positively correlated to V and D , but significantly negatively correlated to TS and TE. BD was closely related to the soil structure and the pore characteristics. The smaller the BD, the larger the number of pores and the better the pore connectivity in the soil. This in turn results in a larger K_s value and a faster transport of solute in the soil (Wen, 2002). SWC was significantly positively correlated to TE and TS. Shipitalo and Edwards (1996) showed that the lower initial water content will result in an earlier initial penetration of solute in the soil characterized by an unstable wetting front. SOC was significantly positively correlated to V and D , but significantly negatively correlated to TS and TE. SOC provides important nutrients for the growth of plants. A greater amount of SOC will promote the growth of plant roots, which facilitates the formation of larger pores. Meanwhile, SOC improves soil structure and thus reduces the resistance of solute transport in the soil. Therefore, it takes less

time for the solute to get transported through the soil with higher SOC content (Zhen et al., 2019). The sand content was significantly positively correlated to V and Pe , but significantly negatively correlated to TS and TE. The opposite trend was observed for silt and clay content. The particle composition reflects the soil texture. A larger sand content will lead to a greater number of macropores in the soil, which favors the formation of preferential flow paths. As a consequence, the solute will be transported faster in the soil. In contrast, silt and clay will generate a greater number of fine pores in the soil, which would increase the diffusion strength and impedes the transport of solute in the soil (Kung et al., 2000).

4. Discussion

The transport behavior of Cl^- differs in soil samples with the same texture but covered by different types of vegetation. For sandy soils covered by different types of vegetation, TS and TE followed the order of S-AR < S-GR < S-SH at a depth between 0 and 40 cm, and S-AR < S-SH < S-GR at a depth between 40 and 100 cm. For loamy soils covered by different types of vegetation, TS and TE followed the order of L-AR < L-SH < L-GR at all different depths in the soil profile. The average V of Cl^- transport in the profiles of the sandy soils followed the order of S-AR (1.67 cm h^{-1}) > S-GR (1.51 cm h^{-1}) > S-SH (1.23 cm h^{-1}), whereas that in the profiles of the loamy soils followed the order of L-AR (1.04 cm h^{-1}) > L-SH (1.03 cm h^{-1}) > L-GR (0.76 cm h^{-1}). Zhen et al. (2015) suggested that diverse root distribution patterns of different vegetation types will result in different soil structure and pores characteristics. The porosity of the soil samples covered by different types of vegetation followed the order of arbor land > shrubland > grassland. Therefore, soil covered by arbor land possesses the best-developed

Table 2

The correlation coefficients between soil properties and Cl^- transport parameters.

Cl^- transport Parameters ^a	BD ^b	SWC ^c	SOC ^d	K_s ^e	TP ^f	Sand	Silt	Clay	Macropore number	Connectivity density
TE	0.787**	0.321*	−0.628**	−0.671**	0.359	−0.659**	0.656**	0.567**	−0.754**	−0.741**
TS	0.870**	0.366*	−0.669**	−0.722**	0.349	−0.432*	0.429*	0.385*	−0.767**	−0.732**
V	−0.956**	−0.270	0.611**	0.894**	−0.376	0.649**	−0.644**	−0.761**	0.773**	0.874**
D	−0.717**	−0.077	0.457*	0.641**	−0.104	0.240	−0.217	0.225	0.370*	0.474*
Pe	0.510	−0.213	0.111	−0.480*	−0.346	0.509**	−0.518**	0.536**	0.396*	0.186

^a TE, TS, V , D and Pe represent initial penetration time, the entire penetration time, the average flow velocity, the hydrodynamic diffusion coefficient, and Péclet number, respectively.

^b BD means bulk density.

^c SWC means soil water content.

^d SOC means soil organic carbon.

^e K_s means saturated hydraulic conductivity.

^f TP means total phosphorus. ** and * mean significant correlation at a level of 0.01 and 0.05, respectively.

preferential flow paths in terms of both quantity and quality, the fastest average flow velocity V of the solute, the smallest TS and TE, and a slightly left-shifted peak value in the BTC (Zhen et al., 2015). These characteristics are consistent with the test results obtained in this study. Furthermore, soil covered by different types of vegetation also exhibit different levels of water infiltration capability. The high infiltration rate found in soil covered by arbor land will further promote solute transport in soil. However, the growth of plant roots is affected by a combination of factors, including the soil texture, nutrient conditions, and SWC. In this study, the root biomass in the sandy soils followed the order of S-AR (277.4 g m^{-2}) > S-GR (131.6 g m^{-2}) > S-SH (112.6 g m^{-2}) at a depth between 0 and 40 cm, and S-AR (84.5 g m^{-2}) > S-SH (77.2 g m^{-2}) > S-GR (40.1 g m^{-2}) at a depth between 40 and 100 cm. The root biomass in the loamy soils followed the order of L-AR > L-SH > L-GR regardless of the soil depth. The total root biomass in the soil samples collected from the 0–100 cm profile followed the order of L-AR (422.5 g m^{-2}) > L-SH (370.2 g m^{-2}) > L-GR (196.5 g m^{-2}). Due to the different distributions of arbor, shrub, and grass roots in the soils with different textures, the number and connectivity density of pores varied substantially between the sandy and loamy soils. Therefore, the transport behavior of Cl^- did not follow exactly the same pattern in sandy soil and loamy soil covered by different vegetation types. Zhang et al. (2004) pointed out that the nutrient loss in soil would be influenced by vegetation restoration through the changes in the pore distribution and preferential flow paths. In addition, they found that the types, forms, and ways of nutrient loss will vary in soils covered by different vegetation types. Therefore, a reasonable vegetation configuration should be adopted to reduce the risk of nutrient loss.

Transport behavior of solute in soil is affected by not only the type of vegetation cover, but also the texture of the soil. When covered by the same vegetation, the loamy soils possessed a higher TS and TE of Cl^- than the sandy soils at all different depths (L-AR > S-AR; L-SH > S-SH; and L-GR > S-GR). At the same time, the average flow velocity V of Cl^- in the sandy soil was greater than that in the loamy soil (S-AR > L-AR; S-SH > L-SH; and S-GR > L-GR). This is because pore distribution in the soil is determined by its mechanical composition. Sandy soil is mainly composed of coarse-grained sand, which results in a greater number of macro-pores along with a high connectivity density. These features favor the formation of preferential flow paths so that the solute experiences less retardation during the transport process. Thus, it takes less time for the solute to accomplish initial penetration and complete replacement. In contrast, the adsorption of ions onto charged aggregates and the complex porous geometry formed by the fine pores in loamy soils will hinder solute transport, thereby resulting in a lower V of Cl^- transport in the soil and a higher TS and TE (Liu and Xu, 2005). Kung et al. (2000) showed that the transport velocity of Cl^- in soil containing a high content of clay particles is much slower than that in soil containing a high content of silt particles. Liu and Xu (2005) analyzed the transport behavior of solute in soil of different textures and found a greater solute transport speed in sandy soil than that in loamy soil. Due to the higher solute transport rate, sandy soil is more prone to loss of nutrients than loamy soil. Through field leaching experiments, Zhang and Fang (2006) found that nutrients will transport with water more easily in sandy soil and migrate to water bodies in a short time. The loss of nutrients often occurs during rainfall after fertilization. Because the sandy soils are very infertile in the studied area, fertilizers were commonly applied when the tree and shrub seedlings were transplanted into the sandy soils. Therefore, high-volume fertilization should be avoided on sandy soil because of the relatively high velocity of solute transport. Instead, the fertilizers should be applied in small quantities over a period of time.

Table 3Pedotransfer function equations of Cl^- transport rate (V) in the studied soils.

Plot ^a	Pedotransfer function equations of V^b	R^2
S-AR	$V = 17.571 - 9.592\text{BD} - 0.720K_s - 0.049\text{Clay}$	0.998
S-SH	$V = 4.919 - 7.776\ln(\text{BD}) - 0.027K_s + 0.108\text{Clay}$	0.822
S-GR	$V = 5.735 - 0.713e^{\text{BD}} + 0.314\ln(K_s) - 0.210\ln(\text{Clay})$	0.986
L-AR	$V = 2.715 - 2.525\text{BD} + 0.365\ln(K_s) + 1.092\ln(\text{Clay})$	0.999
L-SH	$V = 6.403 - 2.914\text{BD} + 0.005\ln(K_s) - 0.557\ln(\text{Clay})$	0.951
L-GR	$V = 3.882 + 0.099e^{\text{BD}} + 0.476\ln(K_s) - 0.067\text{Clay}$	0.997

^a S-AR, S-SH, and S-GR represent *P. sylvestris*, *S. psammophila*, and *A. desertorum* in sandy soils in the Gechougou watershed, respectively; and L-AR, L-SH, and L-GR represent *P. armeniaca*, *C. korshinskii* and *S. bungeana* in loamy soils in the Liudaogou watershed, respectively.

^b V , BD, and K_s represent the average flow velocity of Cl^- , bulk density and saturated hydraulic conductivity, respectively.

Measurement of solute transport parameters through soils is time-consuming, laborious and expensive, thus it is practically impossible for measuring solute transport parameters for different soil types. Therefore, indirect methods for estimating the transport parameters by pedotransfer functions (PTFs) are now advancing. Because the transport parameters of Cl^- in the soils were affected by a combination of different soil properties (Table 2), it is difficult to predict transport parameters of Cl^- using a single factor. Taking the average flow velocity V associated with Cl^- transport as an example, PTFs to estimate transport parameters in soil can be established using multiple regression analysis (Table 3). As shown in Tables 3 and V associated with Cl^- transport under different soil texture and vegetation conditions can be accurately predicted using PTFs ($R^2 = 0.822\text{--}0.999$). This study provided a clear understanding of the characteristics of Cl^- transport and the corresponding influencing factors under different soil texture and vegetation conditions in the water–wind erosion crisscross region of the Loess Plateau. Future studies of nutrient (e.g., nitrogen and phosphorus) or pollutant (e.g., heavy metals) transport in different textural soils of the studied area can provide references for the rational configuration of artificial vegetation with respect to reduce nutrient loss and pollutant migration to water bodies.

5. Conclusions

This study investigated the transport characteristics and their influencing factors of Cl^- in soils with different textures (sandy-S and loamy-L), and covered by different vegetation types (arbor-AR, shrub-SH and grass-GR) in the water–wind erosion crisscross region of the northern Loess Plateau. The V and penetration time associated with Cl^- transport were different in soils with the same texture but covered by different vegetation types, which is caused by the varying distribution of root biomass under different vegetation types that in turn affect the number of macropores, the connectivity density and the preferential flow paths in the soil. Cl^- transport parameters also varied in different textural soils covered by the same vegetation type. This is because the pore size and distribution in soil are significantly affected by soil mechanical composition. Sandy soil is characterized by a large number of macropores and a high connectivity density, both of which favor the formation of preferential flow. In contrast, the complex porous geometry formed by the fine pores in loamy soil and the adsorption of ions onto the charged aggregates will hinder the transport of Cl^- . Pedotransfer functions using readily available soil properties can adequately predict V of Cl^- transport under different conditions of soil texture and vegetation type. Future studies of nutrient (e.g., nitrogen and phosphorus) or pollutant (e.g., heavy metals) transport in different textural soils of the studied area can provide references for the rational configuration of artificial vegetation with

respect to reduce nutrient loss and pollutant migration to water bodies.

Author Credit Statement

Yanwu Pei: Data curation, Visualization, and Writing-Original draft preparation; Laiming Huang: Conceptualization, Resources, Supervision, and Writing-Review & Editing; Danfeng Li: Data curation, and Writing-Original draft preparation. Ming'an Shao: Resources, and Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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